SUPERSTORE SALES ANALYSIS

(Using Excel, PostgreSQL, Power BI)

A Project Report Submitted

By

Md. Adil (2301330540025)

(B. TECH)

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institute of AKTU, Lucknow)

to the Faculty of Biotechnology DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW April, 2026

DECLARATION

I hereby declare that the work presented in this report, entitled “Superstore Sales Analysis”, was carried out by me. I have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute. I have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, and results that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original authors/sources. I affirm that no portion of my work is plagiarised, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, I shall be fully responsible and answerable.

Name: Md. Adil

Roll Number: 2301330540025

B. Tech

(Candidate Signature)

Introduction

The objective of this project is to analyse the Superstore sales dataset using PostgreSQL for data querying and Microsoft Power BI for interactive dashboard visualization.

The project demonstrates how SQL can be used to retrieve business insights from raw data and how Power BI transforms those insights into interactive visualizations that assist in business decision-making.

Objectives

- Analyze sales performance over time.
- Identify high-performing regions.
- Find profitable and loss-making products.
- Identify repeat customers.
- Compare sales across categories and segments.
- Build an interactive dashboard for decision-making.

Dataset Description

Dataset Name: Superstore Sales Dataset

Main Columns

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer Name
- Segment
- Country
- State
- Region
- Product Name
- Category
- Sub-Category
- Sales

- Quantity
- Discount
- Profit

Total Records:
Approximately 9,994

SuperStore_Sales_Dataset.csv

Tools and Technologies

Tools	Description
Ms Excell	Data Cleaning
Database	PostgreSQL
Query Tool	pgAdmin 4
Visualization	Microsoft Power BI
Language	SQL

Database Design

- Database Name: superstore
- Table Name: superstore
- Primary Key: row_id
- Measures Used

I. Sales
II. Profit
III. Quantity
IV. Discount
V. Dimensions Used
VI. Region
VII. State
VIII. Category
IX. Sub-Category
X. Customer Name
XI. Product Name
XII. Order Date
XIII. Ship Mode
XIV. Segment

SQL Queries Used

```
CREATE TABLE SUPERSTORE (
   ROW_ID INT ,
   ORDER_ID VARCHAR(20) NOT NULL,
   ORDER_DATE DATE NOT NULL,
   SHIP_DATE DATE NOT NULL,
   SHIP_MODE VARCHAR(30),
         CUSTOMER_ID VARCHAR(20) NOT NULL,
   CUSTOMER_NAME VARCHAR(100) NOT NULL,
   SEGMENT VARCHAR(30),
   COUNTRY VARCHAR(50) NOT NULL,
   CITY VARCHAR(50) NOT NULL,
   STATES VARCHAR(50) NOT NULL,
   REGION VARCHAR(20),
   PRODUCT_ID VARCHAR(30) NOT NULL,
   CATEGORY VARCHAR(30),
   SUB_CATEGORY VARCHAR(50),
   PRODUCT_NAME VARCHAR(255),
   SALES NUMERIC(10,2),
   QUANTITY INT,
```

PROFIT NUMERIC(10,2),

RETURNS integer,

PAYMENT_MODE VARCHAR(20)

);

SELECT * FROM SUPERSTORE

- --TOTAL_SALES,TOTAL_PROFIT,TOTAL_QUANTITY_SOLD--

  SELECT SUM(SALES) TOTAL_SALES

  FROM SUPERSTORE

- --TOTAL_PROFIT--

  SELECT SUM(PROFIT) TOTAL_PROFIT

  FROM SUPERSTORE

- --TOTAL_QUANTITY_SOLD--

  SELECT PRODUCT_NAME ,SUM(QUANTITY) AS SOLD_QUANTITY

  FROM SUPERSTORE

  GROUP BY PRODUCT_NAME

- --TOP_5_SELLING_PRODUCT--

SELECT PRODUCT_NAME,SUM(QUANTITY) SOLD_QUANTITY

FROM SUPERSTORE

GROUP BY PRODUCT_NAME

ORDER BY SOLD_QUANTITY DESC LIMIT 5

- --RETRIVE_REGION_WITH_HIGH_SALES--

SELECT REGION,SUM(SALES) TOTAL_SALES

FROM SUPERSTORE

GROUP BY REGION

ORDER BY TOTAL_SALES DESC LIMIT 1

- --STATES_WITH_HIGHEST_PROFIT--

SELECT STATES,SUM(PROFIT) AS TOTAL_PROFIT

FROM SUPERSTORE

GROUP BY STATES

ORDER BY TOTAL_PROFIT DESC LIMIT 5

- --CATEGORY_CONTRIBUTES_MOST_REVENUE--

SELECT CATEGORY,SUM(SALES) TOTAL_REVENUE

FROM SUPERSTORE

GROUP BY CATEGORY

ORDER BY TOTAL_REVENUE DESC

- --SUB_CATEGORY_WITH_HIGH_PROFIT--

SELECT SUB_CATEGORY,SUM(PROFIT) TOTAL_PROFIT

FROM SUPERSTORE

GROUP BY SUB_CATEGORY

ORDER BY TOTAL_PROFIT DESC LIMIT 5

SELECT * FROM SUPERSTORE

--CUSTOMER_SPENT_MOST_MONEY--

SELECT CUSTOMER_NAME,SUM(SALES) TOTAL_SPENT

FROM SUPERSTORE

GROUP BY CUSTOMER_NAME

ORDER BY TOTAL_SPENT DESC LIMIT 3

FROM SUPERSTORE

- --ship_mode_used_most_frequently--

SELECT SHIP_MODE,COUNT(ROW_ID) SHIPMENT_COUNT

FROM SUPERSTORE

GROUP BY SHIP_MODE

ORDER BY SHIPMENT_COUNT DESC LIMIT 1

- --Find_the_top_5_customers_by_profit--

SELECT CUSTOMER_NAME, SUM(PROFIT) PROFITS

FROM SUPERSTORE

GROUP BY CUSTOMER_NAME

ORDER BY PROFITS DESC LIMIT 5

- --Which_products_are_making_losses--

SELECT PRODUCT_NAME,PROFIT

FROM SUPERSTORE

WHERE PROFIT<0

- --Compare_sales_and_profit_across_regions--

SELECT SUM(SALES) TOTAL_SALES,SUM(PROFIT) TOTAL_PROFIT

FROM SUPERSTORE

- --Find_monthly_sales_trends--

SELECT SUM(SALES)

FROM SUPERSTORE

WHERE ORDER_DATE BETWEEN '2019-06-1' AND '2019-06-30'

- --Customer_segment_generates_the_most_profit--

SELECT SEGMENT,SUM(PROFIT) PROFIT

FROM SUPERSTORE

GROUP BY SEGMENT

ORDER BY PROFIT DESC

SELECT * FROM SUPERSTORE

- --Repeat_customers--

SELECT CUSTOMER_NAME,count(ORDER_ID)

FROM SUPERSTORE

GROUP BY CUSTOMER_NAME

HAVING COUN

- --States_have_high_sales_but_low_profit--

SELECT STATES, SUM(SALES) SALES,SUM(PROFIT) PROFIT

FROM SUPERSTORE

WHERE PROFIT<0

GROUP BY STATES

ORDER BY SALES DESC LIMIT 1

- --Products_should_be_discontinued_due_to_consistent_losses--

SELECT PRODUCT_NAME,PROFIT

FROM SUPERSTORE

WHERE PROFIT<0

ORDER BY PROFIT ASC LIMIT 5

- --The_best-selling_product_in_each_region--

WITH PRODUCT_SALES AS (

SELECT

REGION,

PRODUCT_NAME,

SUM(SALES) AS TOTAL_SALES,

ROW_NUMBER() OVER (

PARTITION BY REGION

ORDER BY SUM(SALES) DESC

```
) AS RN
FROM SUPERSTORE
GROUP BY REGION, PRODUCT_NAME
)

SELECT
REGION,
PRODUCT_NAME,
TOTAL_SALES
FROM PRODUCT_SALES
WHERE RN = 1
ORDER BY REGION
```

- --3_STATES _WHICH_NEED_FOCUS_FOR_GROWTH--

```
SELECT
STATES,
SUM(SALES) AS TOTAL_SALES,
SUM(PROFIT) AS TOTAL_PROFIT,
ROUND((SUM(PROFIT) * 100.0 / SUM(SALES)), 2) AS PROFIT_MARGIN
FROM SUPERSTORE
GROUP BY STATES
HAVING SUM(SALES) > (
SELECT AVG(TOTAL_SALES)
FROM (
SELECT SUM(SALES) AS TOTAL_SALES
```

```sql
FROM SUPERSTORE
GROUP BY STATES
) T
)
ORDER BY PROFIT_MARGIN ASC
LIMIT 3
```

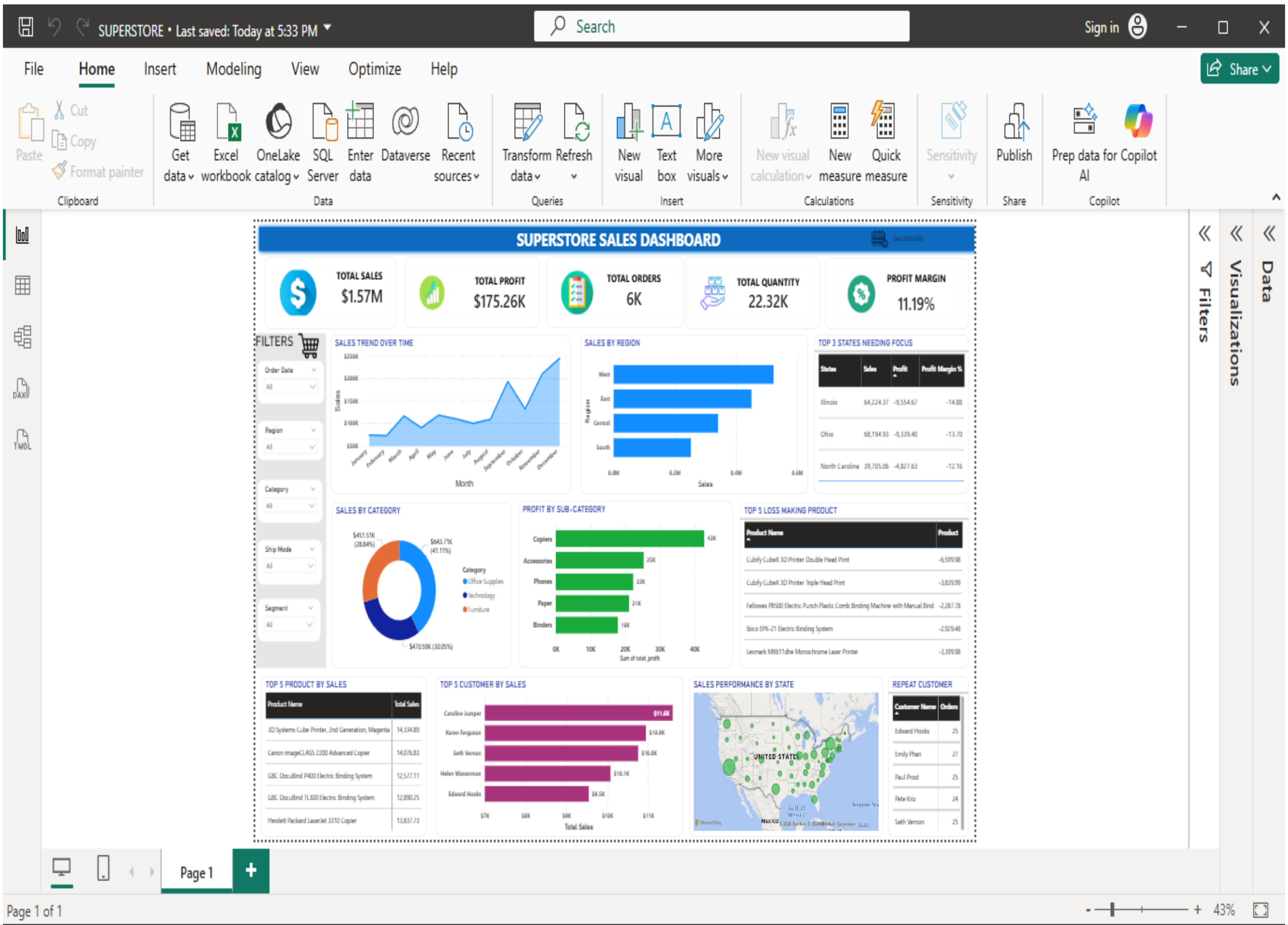
- --Which_states_have_negative_total_profit--

```sql
SELECT STATES ,SUM(PROFIT) AS LOSS
FROM SUPERSTORE
WHERE PROFIT<0
GROUP BY STATES
ORDER BY LOSS ASC LIMIT 1
```

- --sales_trend_over_month--

```sql
SELECT
DATE_TRUNC('month', ORDER_DATE) AS MONTH,
SUM(SALES) AS TOTAL_SALES
FROM SUPERSTORE
GROUP BY DATE_TRUNC('month', ORDER_DATE)
ORDER BY MONTH;
```

Power BI Dashboard



Dashboard Insights

- Technology generated the highest sales.
- The West region recorded the maximum revenue.
- Several states generated high sales but negative profits.
- Canon products contributed significantly to total sales.
- Repeat customers contributed a large share of revenue.
- Certain products consistently generated losses and require pricing review.

Conclusion

This project demonstrates how PostgreSQL and Microsoft Power BI can be integrated for business intelligence and sales analytics. SQL was used to retrieve meaningful business information, while Power BI transformed the results into interactive visualizations. The dashboard helps managers monitor sales, profit, customers, products, and regional performance for better decision-making.